InsightGuard AI

Secure AI-Powered Data Analysis Platform

Project Overview

InsightGuard AI is a Python and Streamlit-based application designed to make basic dataset analysis easier while adding a privacy-awareness layer. Users can upload CSV or Excel files, inspect and clean data, view exploratory charts, receive basic automated observations, ask a limited set of predefined questions, and identify potentially sensitive columns. The application masks detected sensitive columns in its displayed preview and can generate a downloadable PDF summary.

1. Problem Statement

Working with raw datasets often requires several separate steps: checking data quality, cleaning values, exploring patterns, and preparing reports. Datasets may also contain personal information such as names, email addresses, or phone numbers. InsightGuard AI brings several introductory analysis tasks together in one interface and highlights potentially sensitive information before showing the preview.

2. Project Objectives

- Allow users to upload CSV and XLSX datasets.
- Summarize dataset structure, missing values, duplicates, and data types.
- Provide basic cleaning and exploratory data analysis features.
- Generate simple automated insights and answer selected dataset questions.
- Flag potentially sensitive columns and mask them in the displayed preview.
- Generate a downloadable PDF analysis summary.

3. Technology Stack

Technology	Purpose
Python	Core application programming language
Streamlit	Interactive web application interface
Pandas	Dataset loading, cleaning, and analysis
Plotly	Interactive data visualizations
ReportLab	PDF report generation
OpenPyXL	Excel workbook support
python-dotenv	Loading environment variables when configured

4. Main Modules and Implemented Features

Dataset upload and validation: Loads CSV and XLSX files and applies a 10 MB upload-size limit.

Data overview and cleaning: Shows dataset dimensions, missing values, duplicate rows, column details, and basic cleaned data.

Exploratory data analysis: Displays applicable numeric histograms, categorical bar charts, and correlation heatmaps.

Automated insights: Reports basic data-quality observations and selected non-sensitive categorical summaries.

Ask Your Data: Handles a limited set of predefined question patterns, depending on whether the required dataset columns exist.

Sensitive-data detection: Checks column names against sensitive keywords and scans cell values for email and Indian mobile-number patterns.

Masked preview: Creates a display copy and masks detected sensitive columns in the preview.

PDF export: Creates a downloadable report with dataset statistics and a security summary.

5. Security Approach

The current privacy feature uses heuristic detection. Sensitive column names are identified using a keyword list. In addition, regular expressions scan values for email-address patterns and Indian mobile-number patterns, including an optional +91 prefix. When a column is flagged, the application masks its values in the displayed preview copy. This is a useful demonstration of privacy-aware analytics, but it is not a complete PII discovery or data-loss-prevention solution.

6. Testing Performed

Test	Observed result
Upload sample CSV	Successful
Detect sensitive columns by name	Customer_Name and other keyword-matched columns detected
Detect email in a neutral column	Contact_Info detected from email-like values
Detect Indian mobile number in a neutral column	Other_Info detected from mobile-number-like values
Masking in preview	Customer_Name, Contact_Info, and Phone masking observed; phone masking for Other_Info
PDF report generation	Previously generated and downloaded successfully

7. Limitations

- PII detection can miss values that do not match the configured keywords or regular expressions, and may flag unrelated values.
- Phone-number scanning is currently tailored to Indian mobile-number formats.
- Masking applies to the displayed preview; it does not encrypt or anonymize the original uploaded file.
- Ask Your Data supports selected predefined patterns rather than unrestricted natural-language reasoning.
- The project is an MVP and requires additional validation, privacy review, and deployment hardening before real or confidential data is used.

8. Future Enhancements

- Expand PII detection to additional identifiers and configurable regional formats.
- Add automated tests for malformed files, unsupported formats, file-size limits, and masking edge cases.
- Improve privacy controls and secure data handling for deployment.
- Extend question answering and add clearer explanations for generated insights.
- Improve report customization and add a deployment-ready configuration.

9. Conclusion

InsightGuard AI demonstrates how data analysis workflows and basic privacy-aware checks can be combined in a single lightweight application. The current MVP provides file upload, data exploration, selected question answering, sensitive-data flagging, preview masking, and PDF reporting. Further testing and security improvements are recommended before production use.

Prepared as a project overview. Before publishing, add the author's name, institution, project guide, academic year, and repository URL if required.